Quantifying the topic disparity of scientific articles

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Overview

Collaborative research has solved the challenges that our society faces by lowering the barriers between disciplines[1,2]. For this reason, governments and funding organizations design various policies supporting collaborative research within and across institutions[3]. However, it is questionable whether collaborative research is always good for advancing knowledge. Among many aspects that collaborations have, this paper analyzes the extent to which the size of a research team affects the topic disparity of a paper from the overall theme of a discipline. Using document embeddings learned from a natural language model, we show that team size significantly decreases topic disparity in some disciplines, and this tendency is highly correlated with average team size. It implies that large team tends to pursue conventional research topics in these disciplines. We also plan to examine the impact of a research team's diversity, including in our final manuscript.

Data and methods

PLoS One paper data

We parse the XML files for 81,407 research articles published in *PLoS One* after 2016[4] and handled by editors whose name is found in the current list of editors (https://journals.plos.org/plosone/static/editorial-board; accessed on October 1, 2021). We group these articles into 117 disciplines by using editors' section information instead of using discipline keywords given in the XML files. We didn't use the paper's keywords to assign discipline because multiple keywords can be assigned to a single paper, making it hard to identify the primary discipline of a paper.

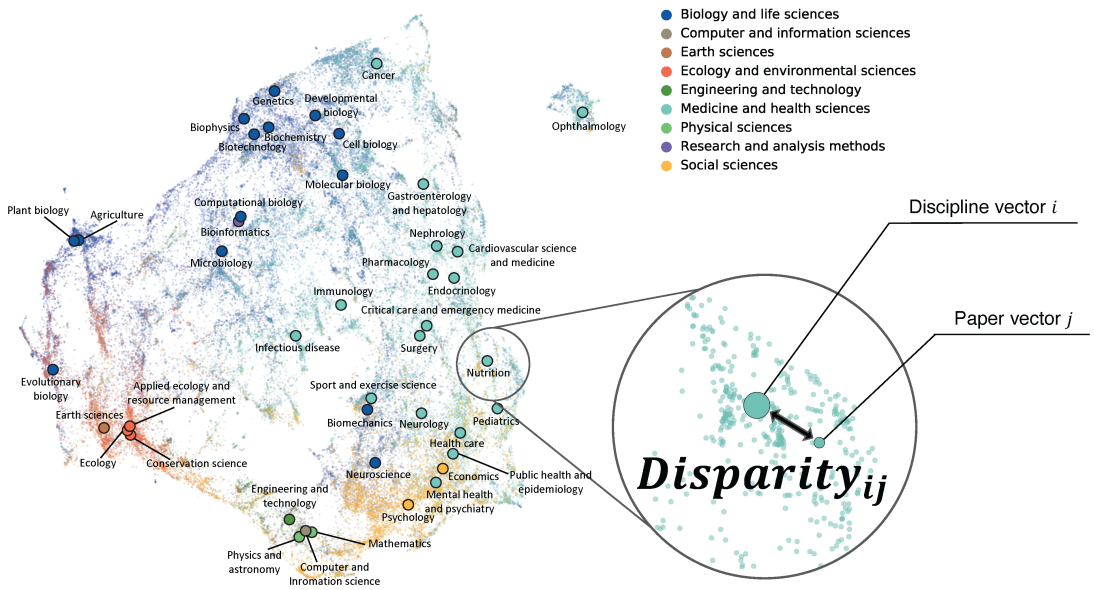

Figure 1. A two-dimensional projection of paper and discipline vectors by UMAP[5]. Each small and large points correspond to a paper and a discipline, respectively. $Disparity_{ij}$ is the cosine distance between discipline i and paper j.

Topic disparity

Next, we convert abstracts and titles of the papers into context-aware vector representation by using SPECTER[6], a pre-trained transformer encoder trained on scientific citations. The mean of paper vectors for a discipline is defined as a discipline vector. Paper vectors and discipline vectors are projected onto a two-dimensional space (Fig. 1) with the UMAP algorithm[5]. Fig. 1 confirms that our approach captures the structure of science disciplines well as it is consistent with existing maps of science[7,8]. Small and large points in (Fig. 1) correspond to papers and the 43 disciplines having more than 1,000 papers, respectively.

The vector representations of papers and disciplines allow us to calculate the topic disparity of a paper (hereafter referred to as the disparity). We define the disparity as the cosine distance between the paper vector and its associated discipline vector. High disparity means that the topic of a paper deviates much from the overall topic of a discipline. For example, a paper[9] that discussed statistical biases in meta-analysis has a disparity value of 0.103 (bottom 0.1%) with the Mathematics vector. On the other hand, an interdisciplinary paper[10] developing a mathematical model of the economic effect of COVID-19 has a disparity value of 0.318 (top 5%) with the Mathematics vector. Then, why do some papers have high disparity values compared to others?

A potential factor is the team size, as people tend to make fewer ideas and avoid risky challenges when they are in a large team[11,12]. In addition, scientific impact and disruptiveness would depend on team work[13,14]. These references suggest the number of authors affects the disparity. We confirm that this assumption holds in many disciplines through OLS regressions,

$$Disparity_{ij} = \beta_{1i} + \beta_{2i} n_{ij} + e_{ij}, \tag{1}$$

where $Disparity_{ij}$ is the disparity of paper j in discipline i; n_{ij} is the number of authors; and β_1 and β_2 are estimated coefficients. Positive β_2 indicates a tendency in a discipline that large teams work on different topics from the overall topic more than small teams. To remove the effect of extremely large teams on this tendency, papers of which the number of authors is larger than the 95th percentile for each discipline are excluded in the regressions.

Result

From the regressions, we find that the number of the authors significantly affects the disparity in 32 out of 43 disciplines ($p < 0.05$; Table. 1). The relationships between the number of authors and disparity are different by discipline. Interestingly, the coefficients of the regression are correlated with the average number of authors of each discipline (Fig 2). A discipline where team size is large on average, mostly in “Biology and Life Sciences” or “Medicine and Health Sciences”, shows a negative relationship between the number of authors and the disparity. On the other hand, disciplines of which the average team size is relatively small tend to have a positive correlation between the number of authors and the disparity. This result shows that if large teams are common in the discipline, small teams tend to study different topics compared to large teams in the same discipline. Similarly, in the disciplines where small teams are common, large teams tend to pursue different topics.

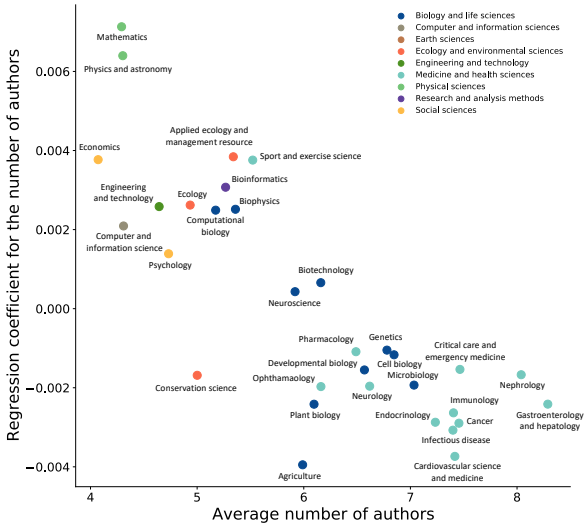

Figure 2. Negative correlation between average numbers of authors and regression coefficients

Table 1. Regression results for the 43 disciplines having more than 1,000 papers.

(a) Disciplines showing significant relationship between the number of authors and topic disparity

Discipline (No. Observations)	Intercept	The number of author	R^2
Agriculture (3288)	$0.24322^{***}(0.0030)$	$-0.003949^{***}(0.00046)$	0.0215
Applied ecology and resource management (990)	$0.18169^{***}(0.0050)$	$0.003842^{***}(0.00086)$	0.0189
Bioinformatics (1270)	$0.22334^{***}(0.0037)$	$0.003071^{***}(0.00062)$	0.0180
Biophysics (1628)	$0.24057^{***}(0.0027)$	$0.002514^{***}(0.00046)$	0.0174
Biotechnology (2508)	$0.26044^{***}(0.0019)$	$0.000658^{*}(0.00028)$	0.0174
Cancer (6493)	$0.25794^{***}(0.0017)$	$-0.002897^{***}(0.00021)$	0.0289
Cardiovascular science and medicine (3898)	$0.25414^{***}(0.0022)$	$-0.003736^{***}(0.00028)$	0.0446
Cell biology (4456)	$0.25392^{***}(0.0017)$	$-0.001166^{***}(0.00023)$	0.0054
Computational biology (1175)	$0.22160^{***}(0.0037)$	$-0.00287^{***}(0.00064)$	0.0121
Computer and information sciences (2469)	$0.24090^{***}(0.0022)$	$0.002092^{***}(0.00047)$	0.0075
Conservation science (1208)	$0.19513^{***}(0.0047)$	$-0.001685^{*}(0.00085)$	0.0024
Critical care and emergency medicine (1457)	$0.21275^{***}(0.0039)$	$-0.001536^{**}(0.00048)$	0.0063
Developmental biology (1640)	$0.25572^{***}(0.0030)$	$-0.0016^{***}(0.00042)$	0.0076
Ecology (4260)	$0.18807^{***}(0.0025)$	$0.0026^{***}(0.00047)$	0.0071
Economics (2231)	$0.21675^{***}(0.0025)$	$0.0038^{***}(0.00055)$	0.0205
Endocrinology (2467)	$0.24540^{***}(0.0030)$	$-0.0029^{***}(0.00039)$	0.0215
Engineering and technology (1872)	$0.24716^{***}(0.0025)$	$0.002582^{***}(0.00048)$	0.0145
Gastroenterology and hepatology (957)	$0.23010^{***}(0.0046)$	$-0.002415^{***}(0.00051)$	0.0220
Genetics (4730)	$0.25658^{***}(0.0018)$	$-0.001048^{***}(0.00023)$	0.0040
Immunology (3429)	$0.25831^{***}(0.0021)$	$-0.002634^{***}(0.00026)$	0.0293
Infectious diseases (7106)	$0.25335^{***}(0.0015)$	$-0.003073^{***}(0.00019)$	0.0353
Mathematics (1180)	$0.21067^{***}(0.0034)$	$0.007130^{***}(0.00072)$	0.0763
Microbiology (5089)	$0.24806^{***}(0.0018)$	$-0.001933^{***}(0.00023)$	0.0135
Nephrology (1178)	$0.20106^{***}(0.0048)$	$-0.001671^{**}(0.00056)$	0.0068
Neurology (1526)	$0.24987^{***}(0.0028)$	$-0.001960^{***}(0.00039)$	0.0155
Neuroscienc (6889)	$0.24254^{***}(0.0013)$	$0.000430^{*}(0.00020)$	0.0005
Ophthalmology (1311)	$0.19398^{***}(0.0060)$	$-0.001971^{*}(0.00090)$	0.0029
Pharmacology (993)	$0.25900^{***}(0.0033)$	$-0.001088^{*}(0.00046)$	0.0045
Physics and astronomy (1140)	$0.21754^{***}(0.0034)$	$0.006398^{***}(0.00071)$	0.0650
Plant biology (2025)	$0.20439^{***}(0.0036)$	$-0.002416^{***}(0.00055)$	0.0091
Psychology (4059)	$0.20581^{***}(0.0019)$	$0.001389^{***}(0.00036)$	0.0035
Sport and exercise science (1130)	$0.15095^{***}(0.0057)$	$0.003758^{***}(0.00095)$	0.0128

(b) Disciplines showing insignificant relationship between the number of authors and topic disparity

Discipline (No. Observations)	Intercept	The number of author	R^2
Biochemistry (3822)	$0.24987^{***}(0.0018)$	$0.000139(0.00026)$	-0.00019
Earth sciences (1417)	$0.19730^{***}(0.0042)$	$0.000323(0.00080)$	-0.00059
Evolutionary biology (2088)	$0.196240^{***}(0.0034)$	$0.000870(0.00056)$	0.00067
Health care (2009)	$0.20749^{***}(0.0033)$	$0.00065(0.00052)$	0.00028
Mental health and psychiatry (1090)	$0.20408^{***}(0.0041)$	$-0.000198(0.00065)$	-0.00083
Molecular biology (2122)	$0.25765^{***}(0.0024)$	$-0.000286(0.00033)$	-0.00012
Nutrition (1412)	$0.23356^{***}(0.0033)$	$-0.00033(0.00049)$	-0.00040
Pediatrics (1026)	$0.24929^{***}(0.0036)$	$-0.00079(0.00050)$	0.0015
Public health and epidemiology (5850)	$0.22606^{***}(0.0017)$	$0.000160(0.00025)$	-0.00010
Surgery (1254)	$0.24583^{***}(0.0038)$	$-0.000519(0.00052)$	-0.000009

Conclusion

Scientific research is becoming more interdisciplinary with increasing team size[13,15]. Herein, we propose a new approach to quantify the topic disparity of a paper from its discipline and explore the relationship between the topic disparity and team size.

Using the PLoS One paper data, we confirm that there are significant relationships between team size and topic disparity in many disciplines. Disciplines with small team sizes show a positive relationship between topic disparity and team size. However, in disciplines where the average team size is large, topic disparity decreases with increasing team size. This result suggests that national governments and funding institutions should design science policies encouraging collaborative research differently by disciplines.

Our method has several advantages for understanding the advancements of science. First, measuring topic disparity with a pre-trained embedding model does not require us to develop a new model and focus on citation links in contrast to cognitive approaches[16]. It can also be used for various analyses, such as assessing the topic diversity of journals, interdisciplinarity of researchers, and performance of research organizations.

Data Availability

The data of PloS One papers are available through the allofplos package

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